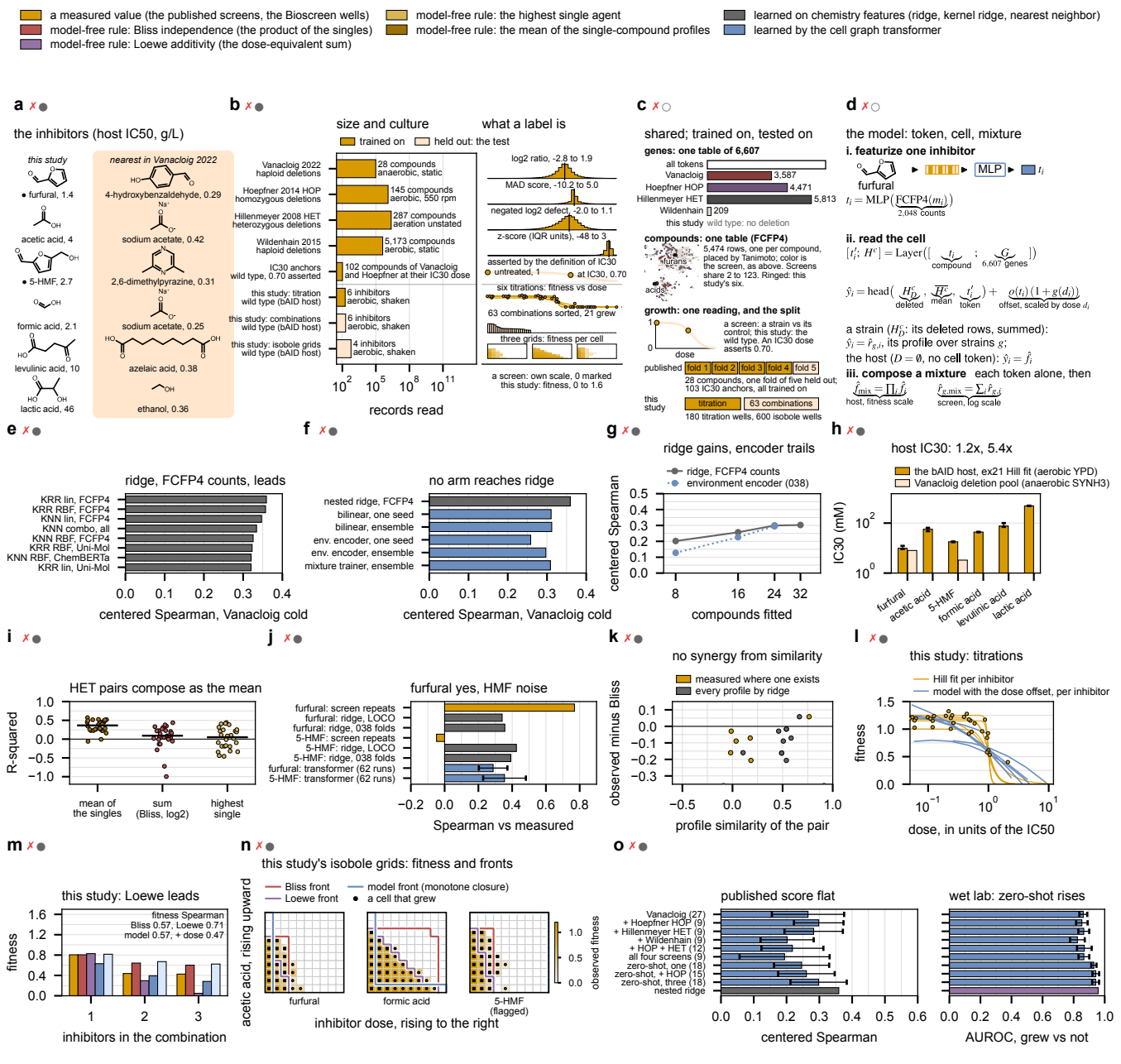

a measured value (the published screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
the inhibitors (host IC50, g/L)
this study
furfural, 1.4
acetic acid, 4
5-HMF, 2.7
formic acid, 2.1
levulinic acid, 10
lactic acid, 46
nearest in Vanacloig 2022
4-hydroxybenzaldehyde, 0.29
sodium acetate, 0.42
2,6-dimethylpyrazine, 0.31
sodium acetate, 0.25
azelaic acid, 0.38
ethanol, 0.36
b
size and culture
trained on
held out: the test
what a label is
Vanacloig 2022 haploid deletions
28 compounds anaerobic, static
Hoepfner 2014 HOP homozygous deletions
145 compounds aerobic, 550 rpm
Hillenmeyer 2008 HET heterozygous deletions
287 compounds aeration unstated
Wildenhain 2015 haploid deletions
5,173 compounds aerobic, static
IC30 anchors wild type, 0.70 asserted
102 compounds of Vanacloig and Hoepfner at their IC30 dose
this study: titration wild type (bAID host)
6 inhibitors aerobic, shaken
this study: combinations wild type (bAID host)
6 inhibitors aerobic, shaken
this study: isobole grids wild type (bAID host)
4 inhibitors aerobic, shaken
records read
log2 ratio, -2.8 to 1.9
MAD score, -10.2 to 5.0
negated log2 defect, -2.0 to 1.1
z-score (IQR units), -48 to 3
asserted by the definition of IC30
untreated, 1
at IC30, 0.70
six titrations: fitness vs dose
63 combinations sorted, 21 grew
three grids: fitness per cell
a screen: own scale, 0 marked
this study: fitness, 0 to 1.6
c
shared; trained on, tested on
genes: one table of 6,607
all tokens
Vanacloig
3,587
Hoepfner HOP
4,471
Hillenmeyer HET
5,813
Wildenhain
209
this study
wild type: no deletion
compounds: one table (FCFP4)
5,474 rows, one per compound, placed by Tanimoto; color is the screen, as above. Screens share 2 to 123. Ringed: this study's six.
furans
acids
growth: one reading, and the split
a screen: a strain vs its control; this study: the wild type. An IC30 dose asserts 0.70.
dose
published
fold 1
fold 2
fold 3
fold 4
fold 5
28 compounds, one fold of five held out; 103 IC30 anchors, all trained on
this study
titration
63 combinations
180 titration wells, 600 isobole wells
d
the model: token, cell, mixture
i. featurize one inhibitor
furfural
MLP
t_i
$t_i = \mathrm{MLP}(\mathrm{FCFP4}(m_i))$
2,048 counts
ii. read the cell
$[t_i'; H^c] = \mathrm{Layer}([\underbrace{t_i}_{\text{compound}}; \underbrace{G}_{6{,}607\ \text{genes}}])$
$\hat{y}_i = \mathrm{head}(\underbrace{H^c_D}_{\text{deleted}}, \underbrace{\bar{H}^c}_{\text{mean}}, \underbrace{t_i'}_{\text{token}}) + \underbrace{o(t_i)(1+g(d_i))}_{\text{offset, scaled by dose } d_i}$
a strain (H^c_D: its deleted rows, summed):
$\hat{y}_i = \hat{r}_{g,i}$, its profile over strains g;
the host ($D = \emptyset$, no cell token): $\hat{y}_i = \hat{f}_i$
iii. compose a mixture each token alone, then
$\hat{f}_{\mathrm{mix}} = \prod_i \hat{f}_i$
host, fitness scale
$\hat{r}_{g,\mathrm{mix}} = \sum_i \hat{r}_{g,i}$
screen, log scale
e
ridge, FCFP4 counts, leads
KRR lin, FCFP4
KRR RBF, FCFP4
KNN lin, FCFP4
KNN combo, all
KNN RBF, FCFP4
KRR RBF, Uni-Mol
KNN RBF, ChemBERTa
KRR lin, Uni-Mol
centered Spearman, Vanacloig cold
f
no arm reaches ridge
nested ridge, FCFP4
bilinear, one seed
bilinear, ensemble
env. encoder, one seed
env. encoder, ensemble
mixture trainer, ensemble
centered Spearman, Vanacloig cold
g
ridge gains, encoder trails
ridge, FCFP4 counts
environment encoder (038)
centered Spearman
compounds fitted
h
host IC30: 1.2x, 5.4x
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
i
HET pairs compose as the mean
R-squared
mean of the singles
sum (Bliss, log2)
highest single
j
furfural yes, HMF noise
furfural: screen repeats
furfural: ridge, LOCO
furfural: ridge, 038 folds
5-HMF: screen repeats
5-HMF: ridge, LOCO
5-HMF: ridge, 038 folds
furfural: transformer (62 runs)
5-HMF: transformer (62 runs)
Spearman vs measured
k
no synergy from similarity
measured where one exists
every profile by ridge
observed minus Bliss
profile similarity of the pair
l
this study: titrations
Hill fit per inhibitor
model with the dose offset, per inhibitor
fitness
dose, in units of the IC50
m
this study: Loewe leads
fitness Spearman
Bliss 0.57, Loewe 0.71
model 0.57, + dose 0.47
fitness
inhibitors in the combination
n
this study's isobole grids: fitness and fronts
Bliss front
Loewe front
model front (monotone closure)
a cell that grew
acetic acid, rising upward
furfural
formic acid
5-HMF (flagged)
observed fitness
inhibitor dose, rising to the right
o
published score flat
Vanacloig (27)
+ Hoepfner HOP (9)
+ Hillenmeyer HET (9)
+ Wildenhain (9)
+ HOP + HET (12)
all four screens (9)
zero-shot, one (18)
zero-shot, + HOP (15)
zero-shot, three (18)
nested ridge
centered Spearman
wet lab: zero-shot rises
AUROC, grew vs not